

Data Preprocessing and Modelling

Introduction

Accurately forecasting sales is an important but difficult process. It requires accurate data that is relevant to sales which can be challenging since multitudinous variables like promotions, location, competition, holidays, etc. have an effect on sales. To predict sales for the Rossman chain in Germany, the following workflow was followed (see Figure 1).

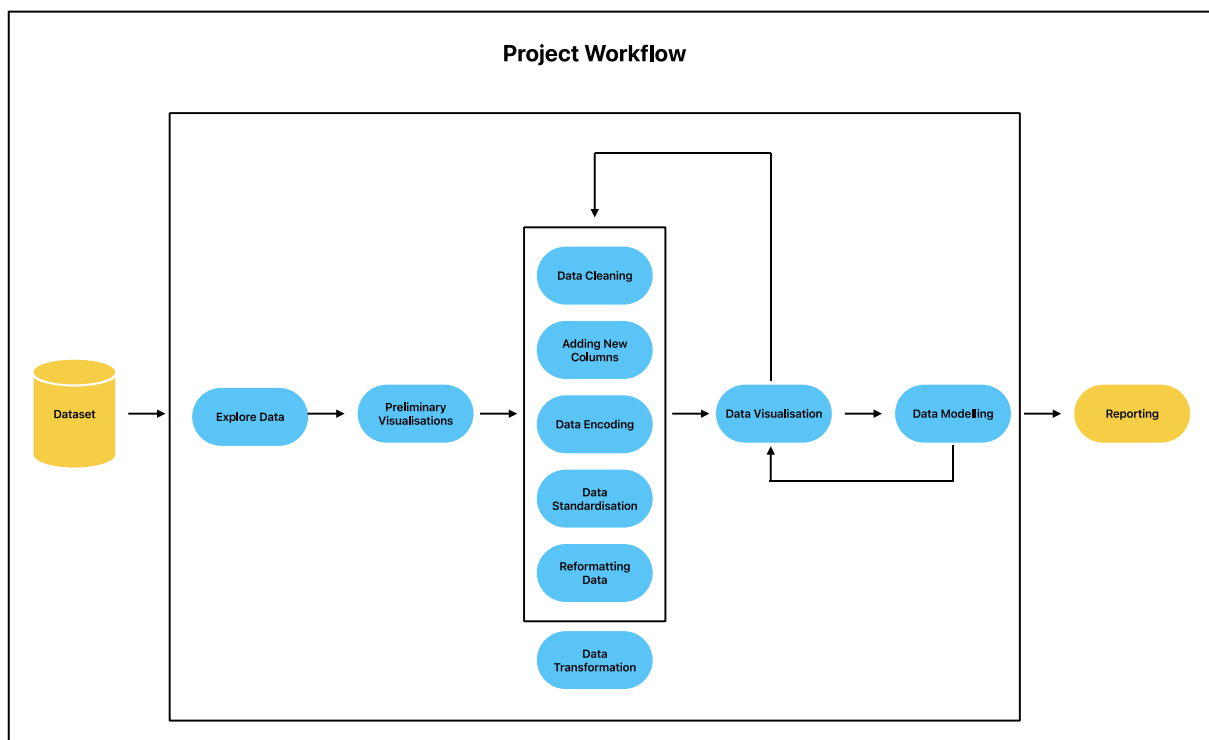


Figure 1: Project Workflow from Pre-Processing to Reporting

Two datasets, namely, store.csv and train.csv were explored. The following tables give us a gist of all the variables present (see Table 1 and Table 2).

Dataset	Field	Description	Example
stores.csv	Store	Anonymized store number	1, 2, 3
	StoreType	Different store models	a, b, c, d
	Assortment	Assortment level	a= basic, b= extra, c= extended
	CompetitionDistance	Distance in meters to the nearest competitor store	1270, 14130, 620
	CompetitionOpenSinceMonth	Approximate month of the time when the nearest competitor was opened	9, 11, 12
	CompetitionOpenSinceYear	Approximate year of the time when the nearest competitor opened	2000, 2007, 2014
	Promo2	A continuous and consecutive promotion	0= store is not participating, 1= store is participating
	Promo2SinceWeek	Calendar week when the store started participating in Promo2	1, 13, 26
	Promo2SinceYear	Year when the store started participating in Promo2	2010, 2012, 2015
	PromoInterval	Consecutive intervals in which Promo2 is re-started	January, April, July, October

Table 1: Variables in Dataset stores.csv

Dataset	Field	Description	Example
train.csv	Store	Anonymized store number	1, 2, 3
	DayOfWeek	Day of week	1= Monday, 2= Tuesday
	Date	Given date	01/01/2013, 31/05/2013
	Sales	Turnover on a given day	46, 667, 889
	Customers	Number of customers on a given day	0, 129, 1061
	Open	An indicator for whether the store was open on that day	0= closed, 1= open
	Promo	Whether a store is running a store-specific promo on that day	0= no promo, 1= promo running
	StateHoliday	Indicates a state holiday	a= public holiday, b= easter holiday, c= Christmas, 0= none
	SchoolHoliday	Indicates if the store was affected by the closure of public schools	0= unaffected, 1= affected

Table 2: Variables in Dataset train.csv

Store.csv provides us with data for 1,115 drug stores with information about their competitors and promotions. Train.csv contains historical sales data covering sales from January 2013 to July 2015. It specifies the sales, customers, holidays, operational status and more for the given stores. These were processed and merged into a single dataset, df (see Table 3), which was used to build a forecasting model to predict sales. With variables like competitors, promotions, holidays, operational status, and store type in one dataset, we can observe their relevance to sales forecasting.

Dataset	Field	Description	Example
df	Store	Anonymized store number	1, 2, 3
	CompetitionDistance	Distance in meters to the nearest competitor store (standardized)	-0.53
	Promo2	A continuous and consecutive promotion	0= store is not participating, 1= store is participating
	StoreType_a	Store model 'a'	0= not present, 1= present
	StoreType_b	Store model 'b'	0= not present, 1= present
	StoreType_c	Store model 'c'	0= not present, 1= present
	StoreType_d	Store model 'd'	0= not present, 1= present
	Assortment_a	Basic assortment level	0= not present, 1= present
	Assortment_b	Extra assortment level	0= not present, 1= present
	Assortment_c	Extended assortment level	0= not present, 1= present
	Promo2SinceDate	The calendar date when the store started participating in Promo2	2014-09-25, 2013-04-13
	PromoInterval_Feb,May,Aug,Nov	Consecutive interval in which Promo2 is re-started	0= not present, 1= present
	PromoInterval_Jan,Apr,Jul,Oct	Consecutive interval in which Promo2 is re-started	0= not present, 1= present
	PromoInterval_Mar,Jun,Sept,Dec	Consecutive interval in which Promo2 is re-started	0= not present, 1= present
	Date	The given date	2015-07-31, 2014-10-28
	Sales	The turnover on a given day	5263, 5020, 4782
	Customers	The number of customers on a given day	555, 546, 523
	Open	An indicator for whether the store was open on that day	0= closed, 1= open
	Promo	Whether a store is running a store-specific promo on that day	0= no promo, 1= promo running
	SchoolHoliday	Indicates if the store was affected by the closure of public schools	0= unaffected, 1= affected
	DayType	Day of the week	Sunday or Other Days
	StateHoliday_0	Indicates no state holiday	0= not present, 1= present
	StateHoliday_a	Indicates a public holiday	0= not present, 1= present
	StateHoliday_b	Indicates Easter holiday	0= not present, 1= present
	StateHoliday_c	Indicates Christmas holiday	0= not present, 1= present
	CompetitionOpenedBefore	If the competition opened before the current store sale dates (standardized)	0.144 or 0 if there is no competition or the competition opened after the sales date.

Table 3: Variables in Dataset df

Visualizations were constructed at three different stages- during preliminary processing, data transformation, and after data modelling. The figure below shows us what exactly we wish to predict with this data, i.e., the remaining sales for 2015.

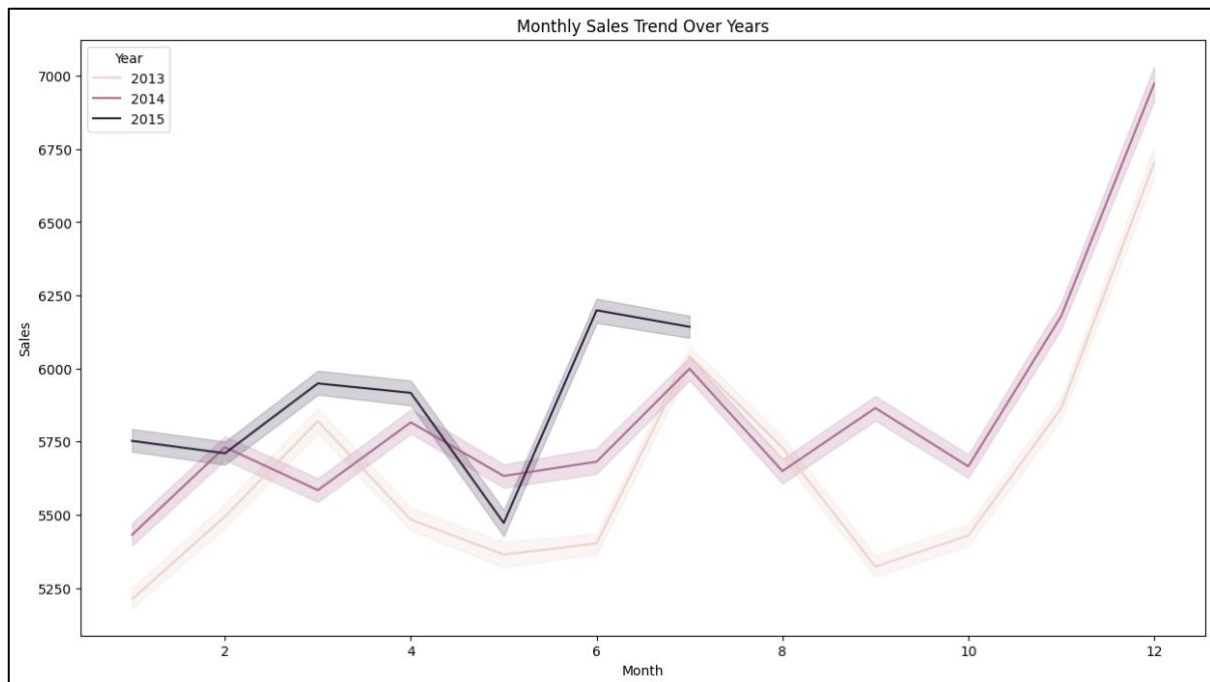


Figure 2: Monthly Sale Trends Over Years (2013,2014,2015)

Ensuring data reliability is a crucial step before analysis. The quality of data can be assessed by considering its accuracy, completeness, consistency, validity, uniqueness, and timeliness. For this project, missing or inconsistent data, duplicate entries, and invalid formatting are key quality issues that need addressing. Failure to do so can lead to inaccurate predictions, affecting the company's revenue and strategic decisions. All these points were kept in mind while working with the merged dataset and the key steps taken are explained in the next section.

Methodology

To understand what aspects of the data need to be cleaned and processed, exploratory data analysis was conducted. This included exploring the data through visualizations like heatmaps, bar charts, line plots, and scatter plots through which we were able to see what the raw data looks like and what it can tell us at the preliminary stage. These visualizations then guided the major preprocessing tasks.

Initially, the variables with multiple values (like store type- a,b,c,d) were visualized to see the distribution of the types. Bar charts are very simple and straightforward visualizations to use when working with frequencies. Therefore, the following two bar charts were made for Stores.csv (see Figure 3 and 4).

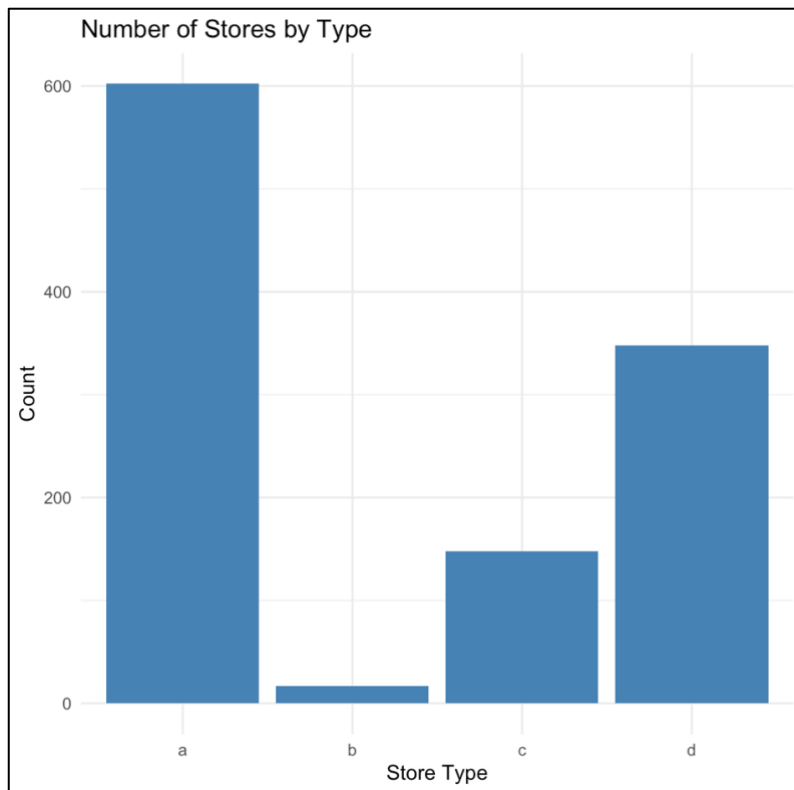


Figure 3: Bar Graph of Number of Stores by StoreType

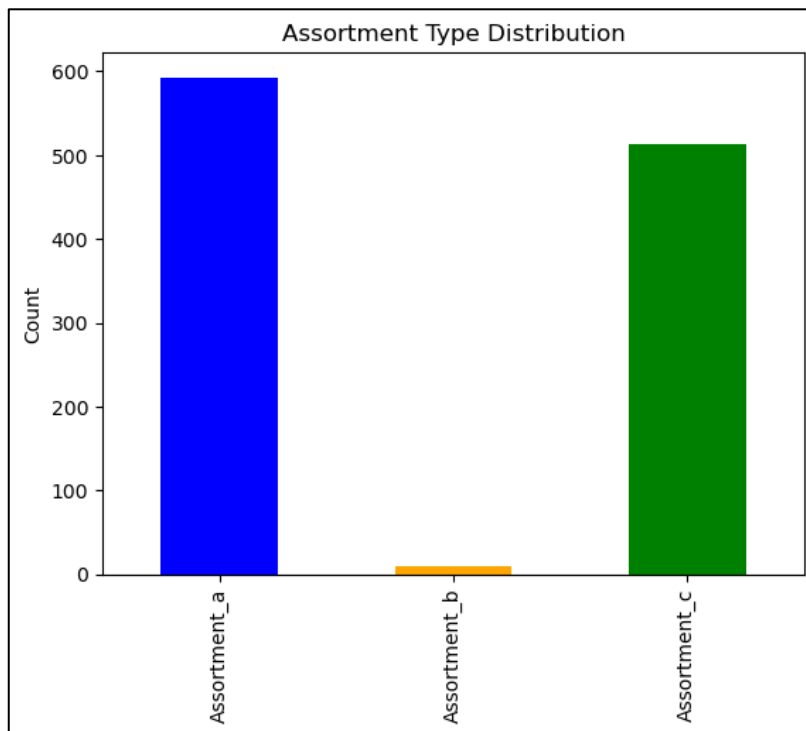


Figure 4: Bar Graph of Number of Stores by Assortment Type

For PromoInterval, a treemap was constructed with Sales (see Figure 5). A treemap is an easy way to display data that is grouped together.



Figure 5: Treemap of Sales by PromoInterval

Since the spread of these values is very diverse, these categorical features were converted into numerical formats using one-hot encoding. This allows us to conduct in depth analysis with these variables.

Addressing missing data and duplicate entries is extremely importance; the next step included making a heatmap with all the present null values in the dataset stores.csv (see Figure 6). A heatmap shows the magnitude of values of interest using colors.

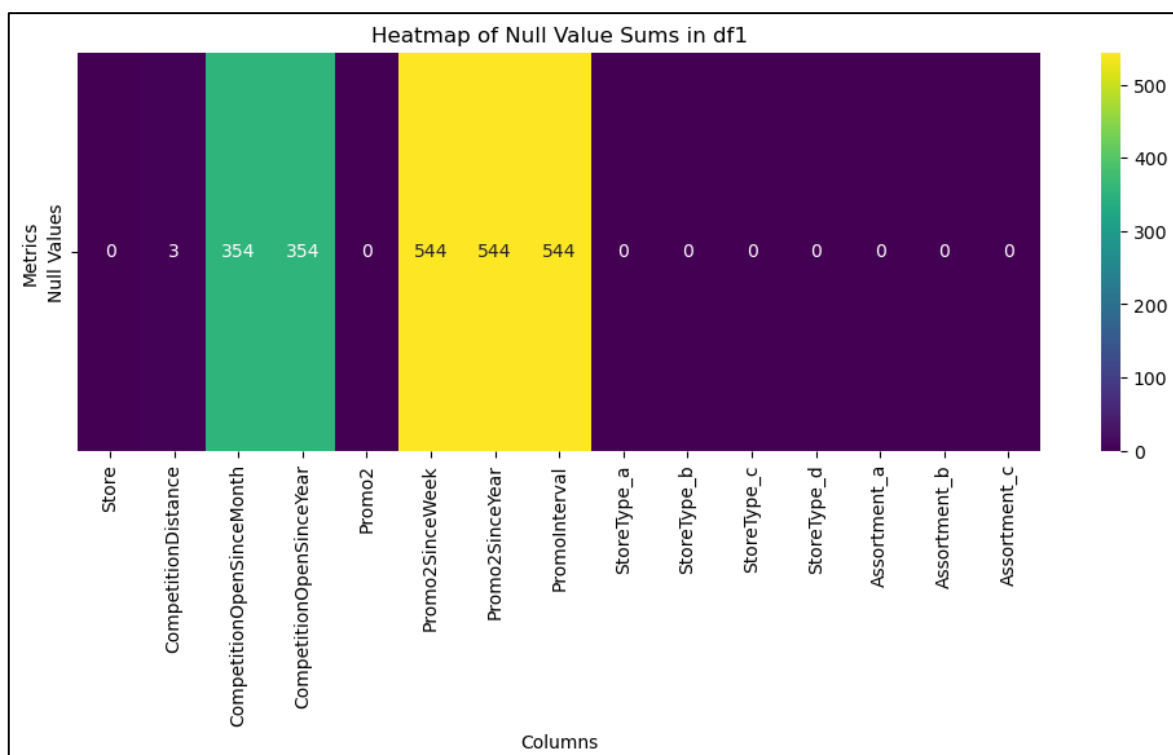


Figure 6: Heatmap of Null Values in Stores.csv

This visualization enabled us to make decisions regarding null values. The columns of Promo2SinceWeek and Promo2SinceYear were merged to make Promo2SinceDate. A similar

approach was taken with CompetitionOpenSinceMonth/Year. The null values in Promo2SinceDate were removed by exchanging them with 'No Promotion'. Similarly, 'No Date since No Competition' was used for CompetitionOpenSinceDate. These columns (along with 'Date' in Train.csv) were changed into the format of datetime object as well to ensure consistency. It was not possible to simply delete the null values as they included data for almost half the stores in the dataset. Moreover, these null values also conveyed some information about the store. We were then left with the following heatmap (see Figure 7).

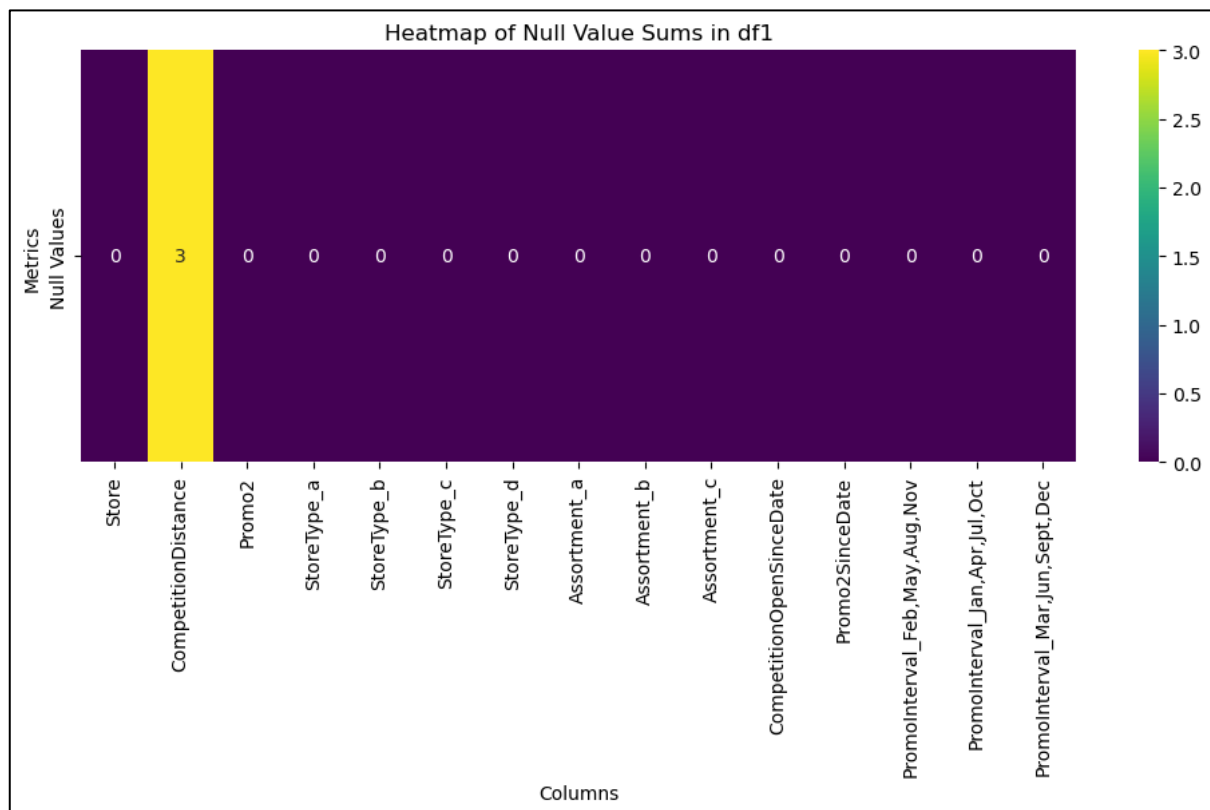


Figure 7: Heatmap of Null Values in Stores.csv

The three values in CompetitionDistance were dropped. Train.csv followed a similar procedure and we saw no null values in the dataset (see Figure 8).

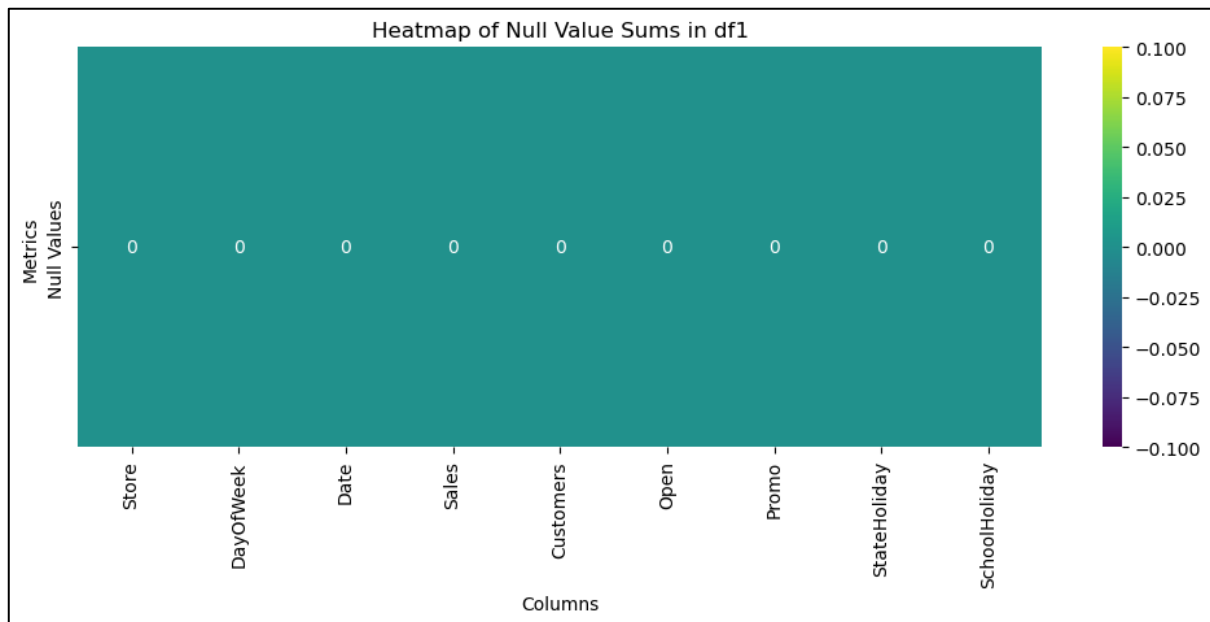


Figure 8: Heatmap of Null Values in Train.csv

The DayOfWeek column in train.csv showed the following spread (see Figure 9).

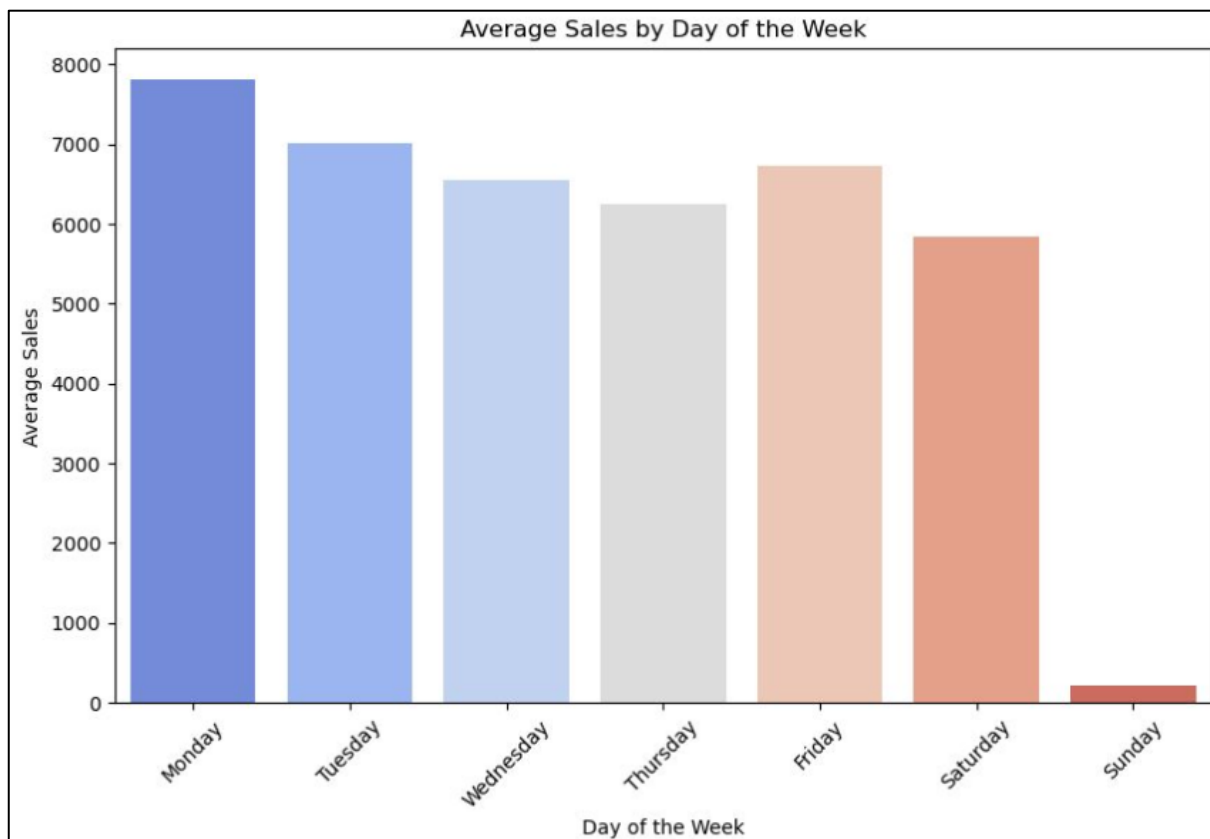


Figure 9: Bar Graph of Average Sales by DayOfWeek

This uneven spread led us to split the values as 'Sunday' and 'Other Days' in a new column called 'DayType' (while dropping the columns of DayOfWeek). Duplicate values were

observed in the column of StateHoliday which was then rectified by combining both into one value (see Figure 10 and 11).

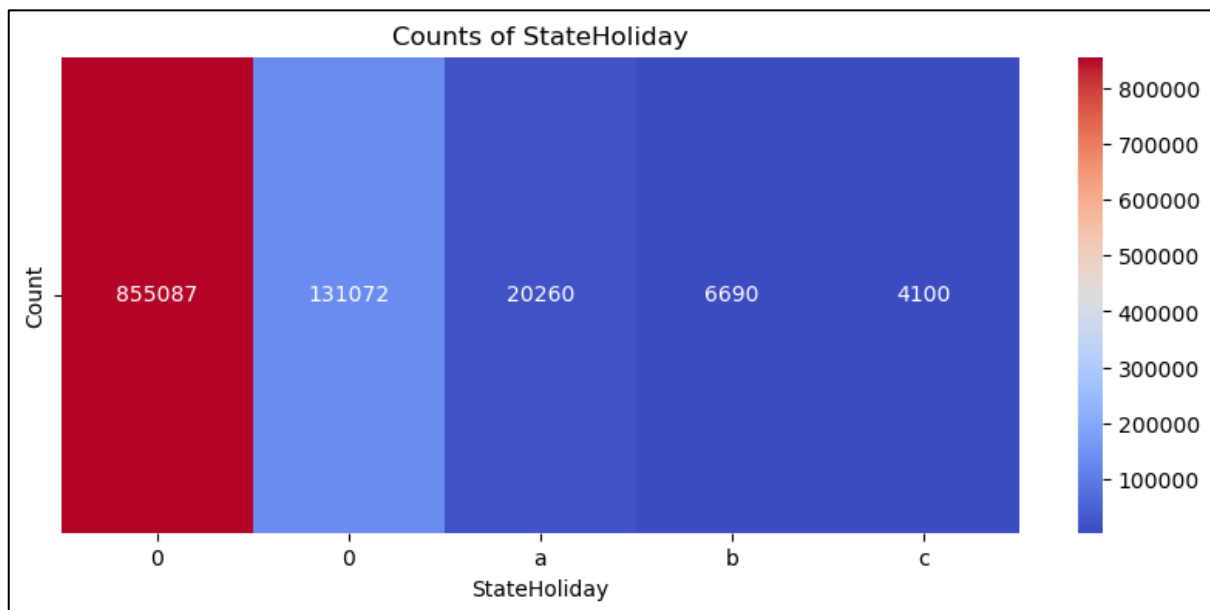


Figure 10: Heatmap of Counts of StateHoliday (showing duplicate values '0')

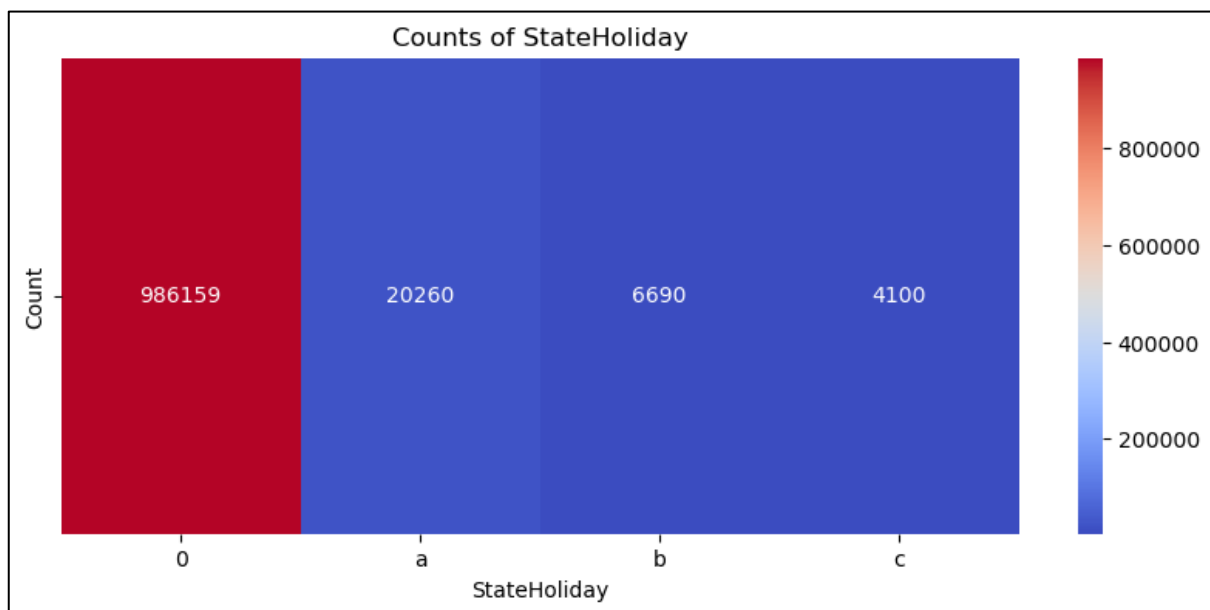


Figure 11: Heatmap of Counts of StateHoliday

These two datasets were then combined to form df. A new column 'CompetitionOpenedBefore' was added which showed us whether the competition opened before the current Sales date. This, along with CompetitionDistance was normalized. This process eliminates bias from magnitude and improves model convergence which was required by the following section.

Modelling

A Random Forest model was created to forecast sales. This model was chosen because sales data often exhibits non-linear relationships due to seasonality, promotions, and holidays, which Random Forest can model effectively. It captures feature importance which helps in identifying the most influential variables and is scalable for large datasets making it suitable for this dataset. To understand the key variables that need to be included in this model, the following visualizations were drawn.

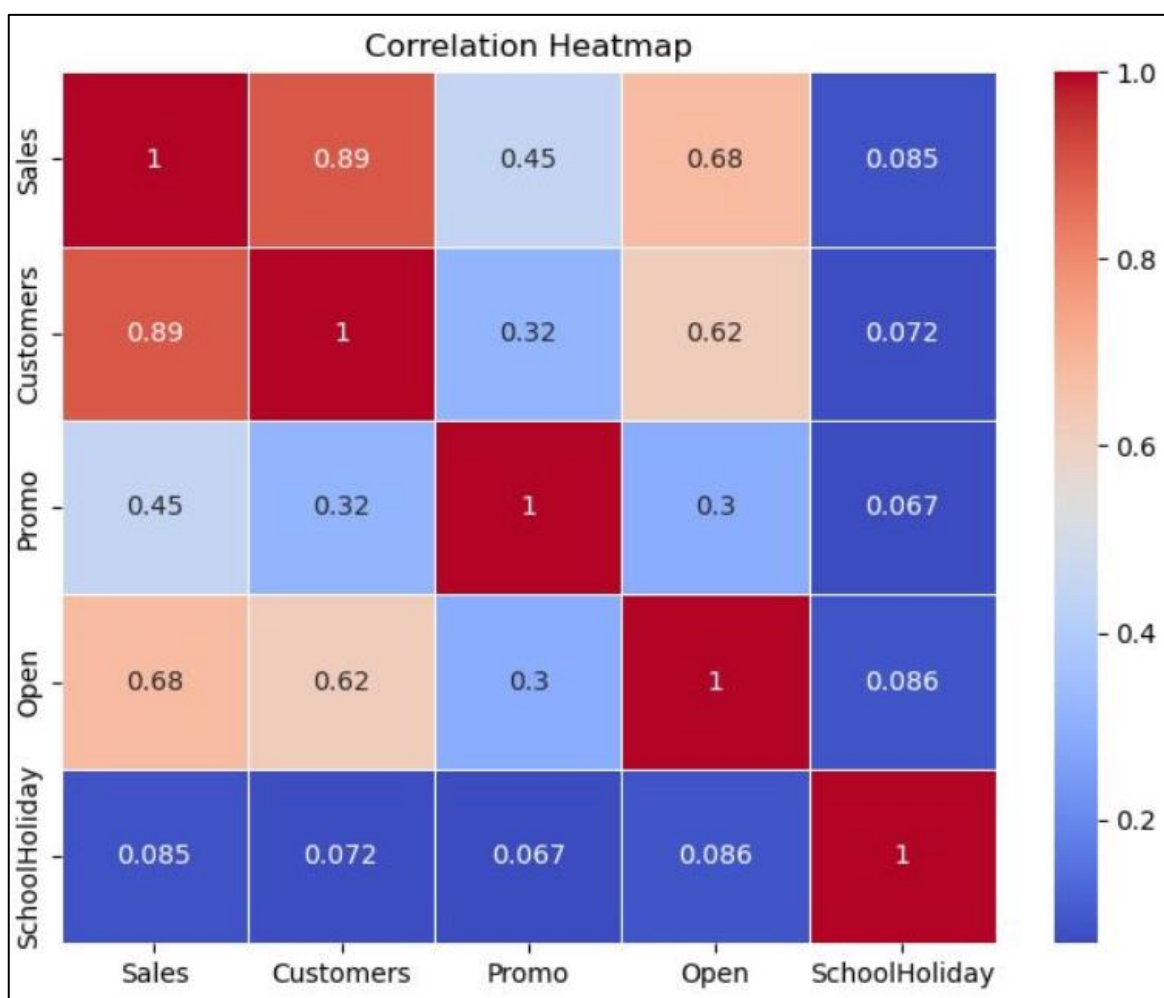


Figure 12: Correlation Heatmap with Variables- Sales, Customers, Promo, Open, SchoolHoliday

A correlation heatmap was generated as it can easily convey correlation between variables through numbers and colors. It is evident that Customers and Sales has the highest correlation (0.89). This is important to know so that we can remove customers from our model to allow

other variables to show their effect on Sales. Customers are also something that the company does not have direct control over as opposed to variables like promotions. Therefore, our first conclusion is to not include Customers in our forecasting model.

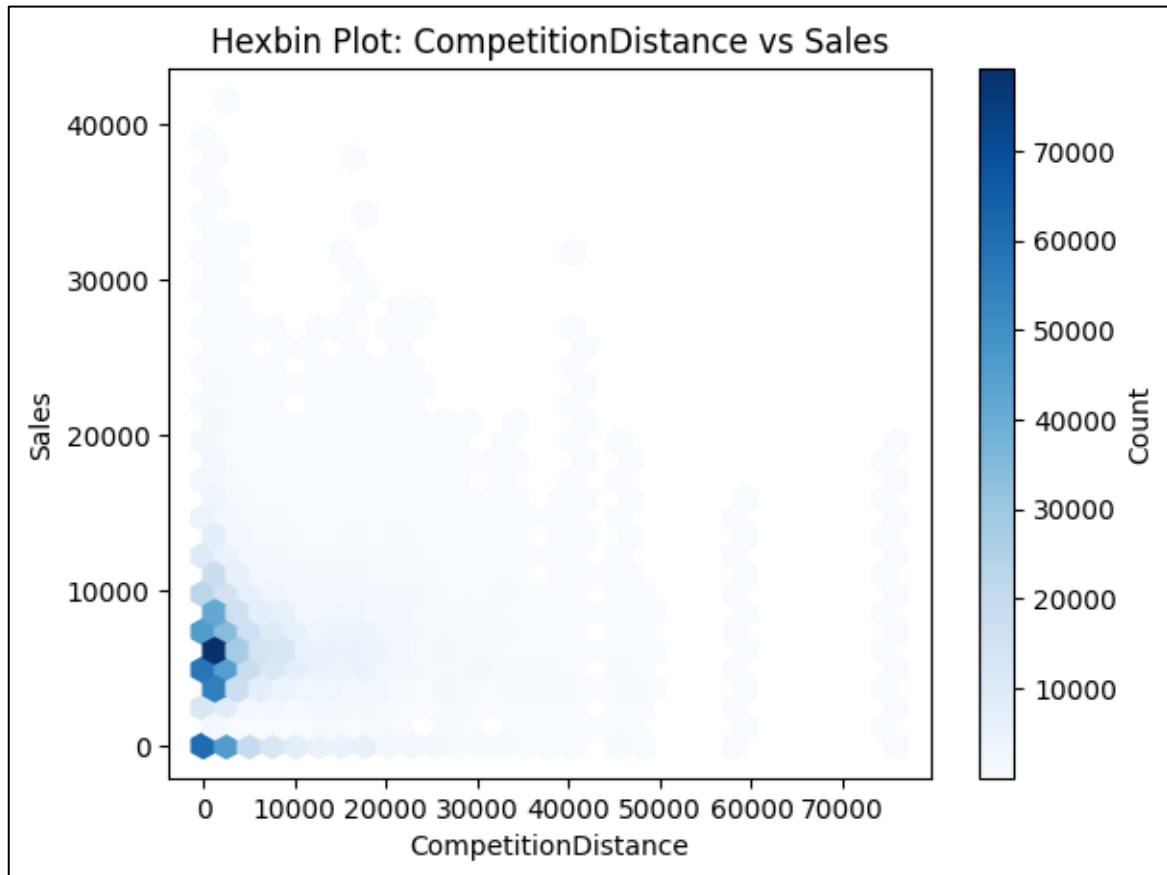


Figure 13: Hexbin Plot for CompetitionDistance and Sales

A hexbin plot is useful when you have two numerical variables with many data points. It uses color to show the frequency of points. Through the above figure, we can see that closer competitors lead to lower number of sales. The dark blue color of the hex explains that a majority of the stores in the dataset lie in this area. Therefore, our second conclusion is that CompetitionDistance has an influence of sales and should be present in the model.

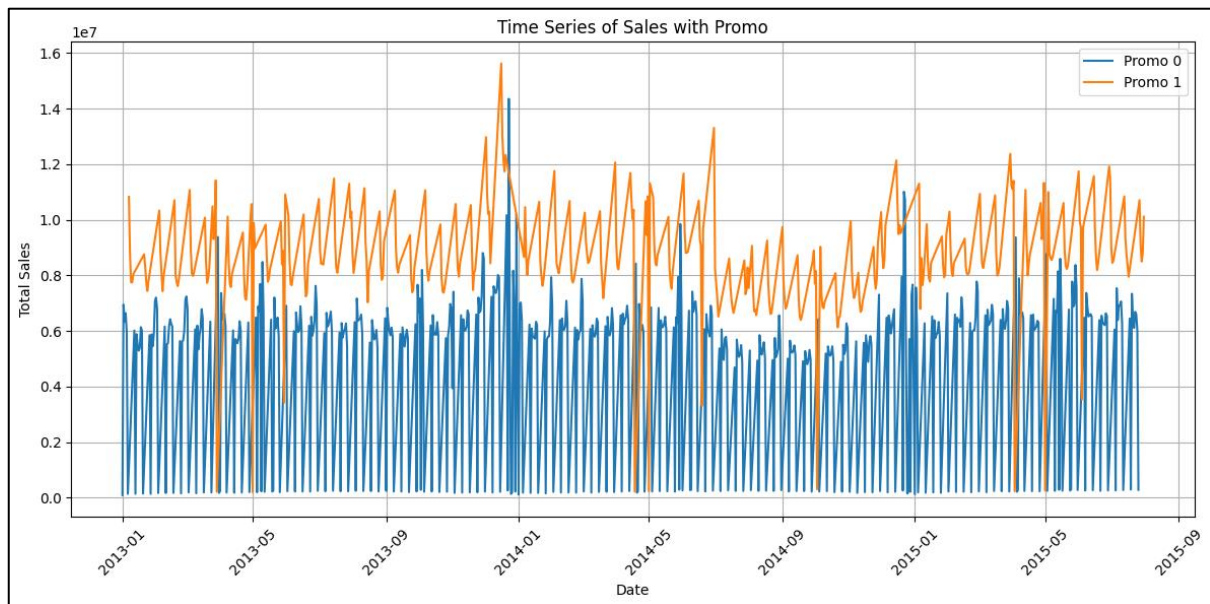


Figure 14: Time Series of Sales with Promo

A time series graph allows us to see the trend of changes in sales over a period of two years for stores that ran a promo (promo=1) and those that did not (promo=0). It is quite evident that running a promo has consecutively lead to higher sales. Therefore, our third conclusion is that Promo influences sales and should be included in the model.

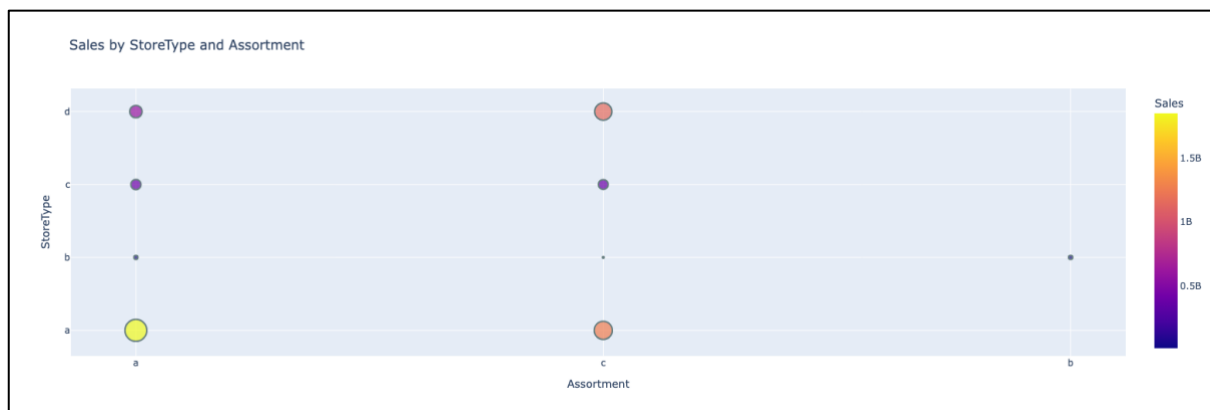


Figure 15: Bubble Chart of Sales by StoreType and Assortment

A bubble chart is similar to a scatterplot; however, each point is assigned a label or category. By the graph above we can see that both StoreType and Assortment show variation in sales and are therefore influential. Our fourth conclusion is that these variables should also be included in the model. Another graph that shows the impact of StoreType on sales is attached below.

With the above conclusions and incorporating other variables, a random forest model was made with Mean Squared Error of 1,936,024.91 and R-squared value of 0.869. The following bar graph shows us the feature importance of all the variables used.

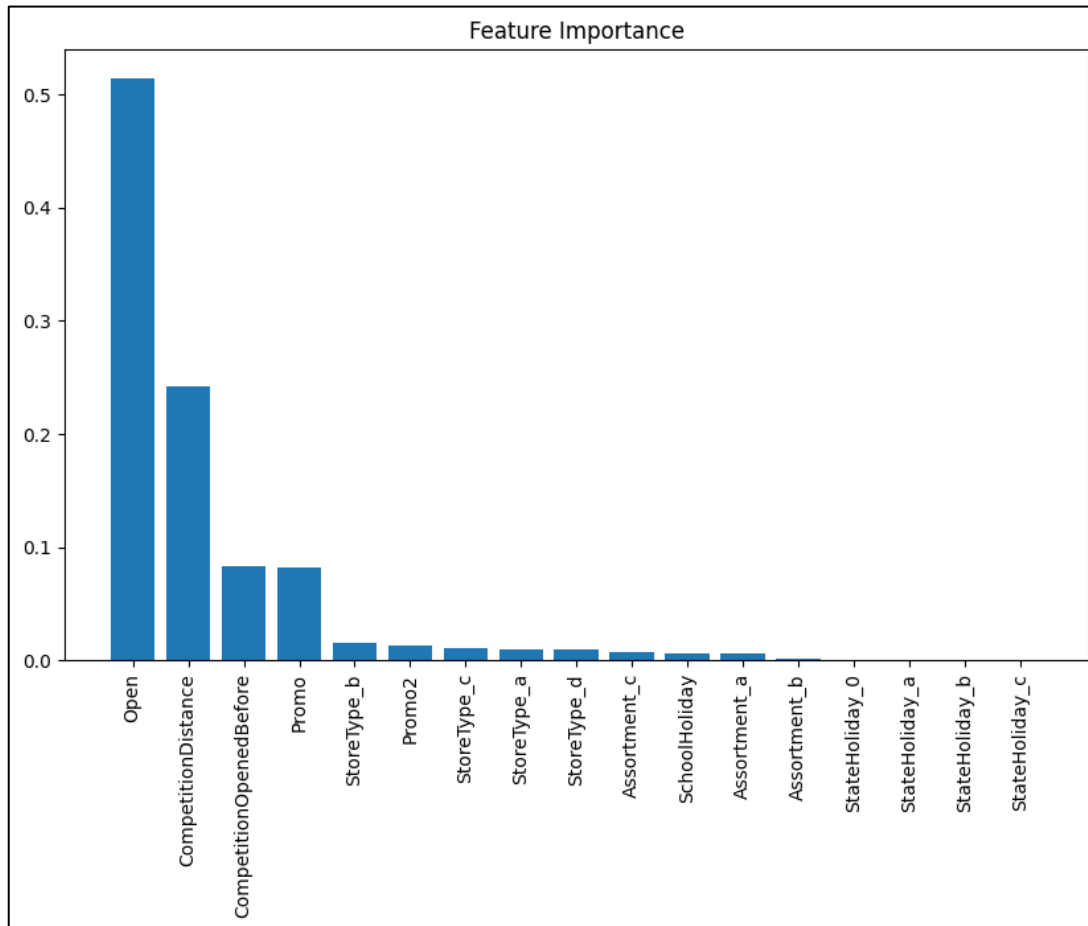


Figure 16: Bar Graph of Feature Importance for Random Forest Regression

Results

The results of the random forest regression model tell us that 86.9% of the variation in sales can be explained by the selected features above. The model includes all key features and has a low MSE score as well, which gives us the perfect balance of predictive power and feature richness. The performance of this model can also be understood by the following graphs.

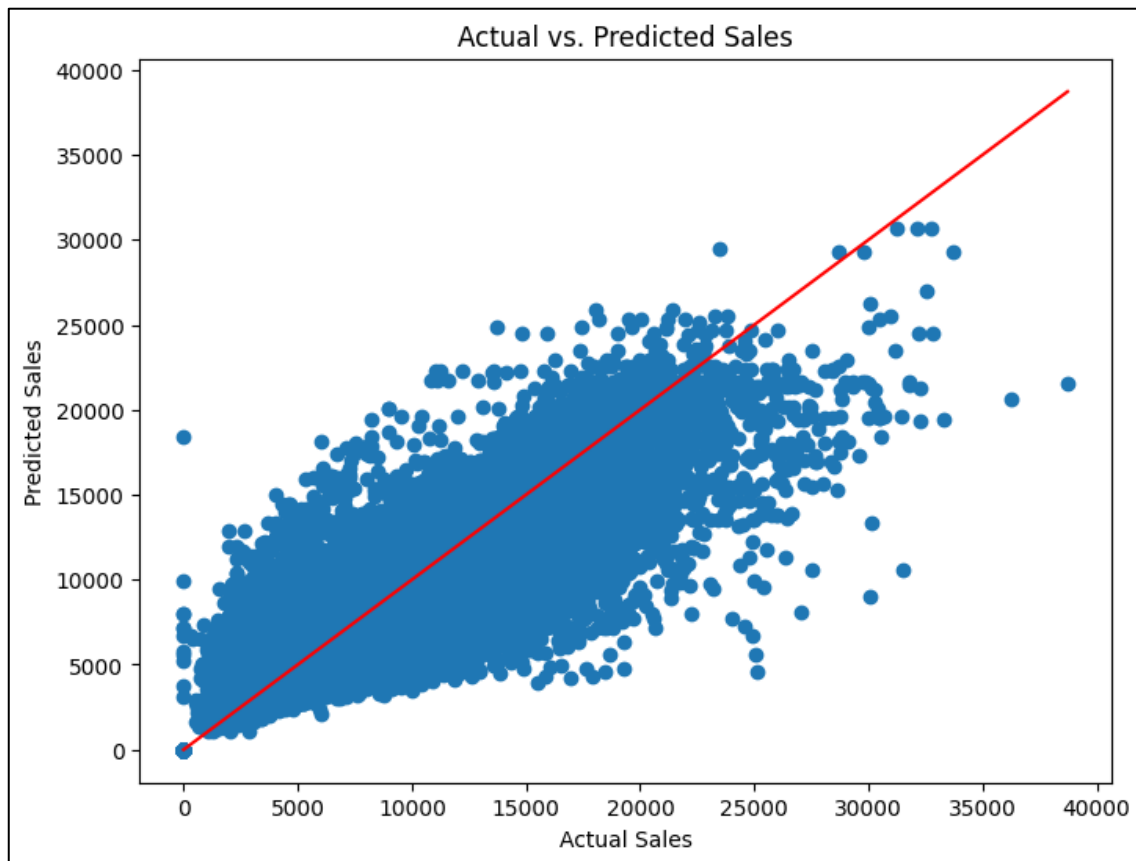


Figure 17: Scatter Plot of Actual vs Predicted Sales

A scatterplot can be used to visualize how close the models predicted values are to the actual values. Since all the points roughly follow the 45-degree line, we can assume that the model is performing well.

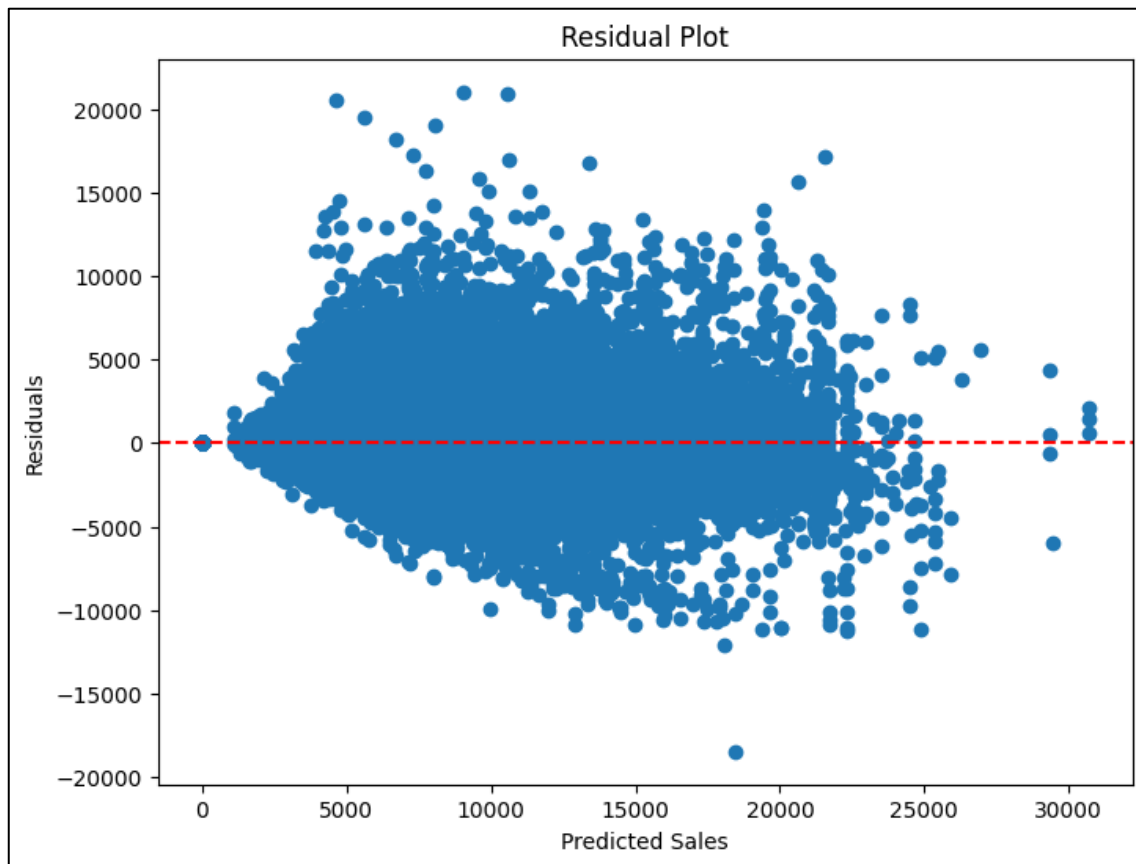


Figure 18: Residual Plot for Predicted Sales

A residual plot helps us check if there are any patterns in the residuals (the difference between the predicted values and the actual values, i.e., errors). The spread of the residual points around zero show us that the model is well-fitted.

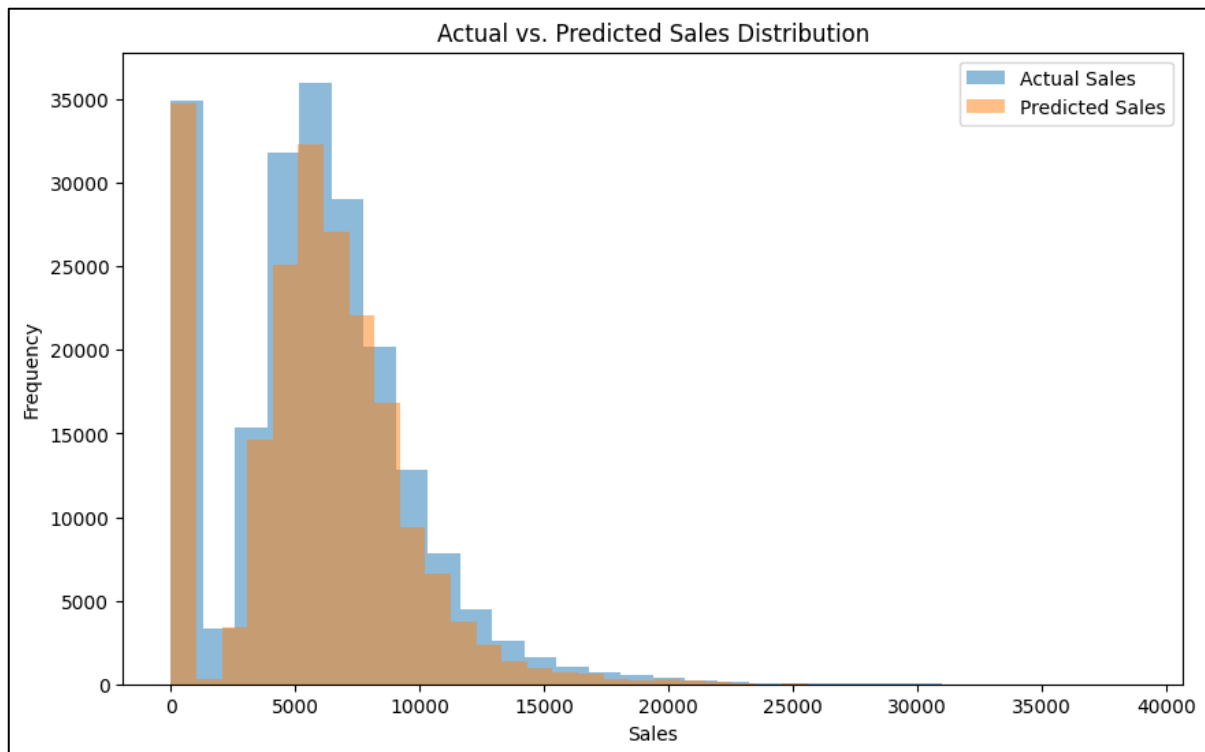


Figure 19: Histogram for Actual vs Predicted Sales

Lastly, this graph shows us that the predicted values follow a similar distribution as the actual values, and therefore, the model is fitted well.

This model benefits from a comprehensive set of features that address all potential factors influencing sales like `CompetitionDistance`, which captures the impact of competition on sales, and `Promo` and `Promo2`, which reflect the effects of short-term and long-term promotional campaigns. The model also includes `StoreType` and `Assortment`, helping to differentiate between stores, while `StateHoliday` captures seasonal sales fluctuations. From the feature importance graph (figure 15), we can see that the operational status of the store ('Open') along with competitor variables, promo, and store type contribute the most towards the model's decision-making process. With the low MSE and high R-squared scores, we can assume that this model will generalize well to unseen data.

Conclusion

We started off this report by exploring two datasets that held raw data about the sales of a company in Germany. We are ending this report with a model that accurately predicts sales for unseen data. Through this report, we saw the lifecycle of raw data from preprocessing to

modelling and the role data visualization plays at every step. The results of this report offer significant business insights by identifying impactful promotional strategies, highlighting the influence of competition on sales, and accounting for seasonal fluctuations, enabling retailers to optimize staffing, inventory, and promotional planning. However, there are certain limitations. Firstly, random forest model is not the best for pure time-series modelling. It also requires substantial computational resources. Secondly, using historical data with constrained variables limits its adaptability to rapidly changing market dynamics or unobserved trends, and while its linear assumptions simplify relationships, they may fail to capture complex interactions between features. For future research, a hybrid approach to modelling is recommended like combining random forest with a time-series model (like LSTM). Moreover, relevant and timely data should be used which may focus on specific regional areas.